

April 2, 2025

California Fish and Game Commission
715 P Street
PO Box 944209
Sacramento, CA 94244

Re: Comments for Western Joshua Tree Conservation Plan

Dear Commissioners,

We are concerned scientists affiliated with the Ecology and Evolutionary Biology department at the University of California, Santa Cruz. We have expertise in California ecosystems and climate change. We appreciate the tremendous effort that has gone into the draft Western Joshua Tree Conservation Plan, and we urge the commission to consider the following comments.

1. Incorporate more realistic emissions scenarios to predict regional climate refugia

The IPCC's low-emissions scenario (SSP 2-4.5) is an overly optimistic projection. At present, the plan defines refugia as areas which will "continue to provide suitable habitat conditions...based on the low emissions scenario," and builds much of its prioritization scheme around these projections. For effective planning, the more probable moderate and high emission scenarios (SSP 3-7.0 and SSP 5-8.5) should be considered as well.

2. Prioritization should not automatically exclude areas with projected climatically suitable habitat loss

The draft plan gives lower priority to populations in areas projected to lose climatically suitable habitat under the low-emissions scenario – areas also frequently under high threats from development (WJTCP report p. 4-33, 4-40). However, populations within these areas may still be valuable and worth prioritizing. Consider the following examples:

a) Local refugia: While the current plan defines refugia as the climatically suitable habitat projected to remain within the existing range, a broader definition of refugia (e.g., McLaughlin et al. 2017) includes local environments *within areas projected to lose climatically suitable habitat* that may remain suitable for small populations because of landscape features that ameliorate water stress and temperatures (e.g., persistent groundwater availability). These local refugia can provide important remnant habitats.

b) Genetic resources: Trailing-edge populations (located within areas of projected climatically suitable habitat loss) could harbor high genetic variation and climate change-adaptive alleles (Rehfeldt et al. 2002; Matías et al. 2019; Klockow et al. 2020, Hampe and Petit 2005). Because these individuals are at the driest and warmest parts of their range, they may contain the genetic resources needed to help the species adapt to climate change in other parts of the range. Maintaining these populations could maintain the species' evolutionary potential.

c) Healthy adult stands: The western Joshua tree has a long lifespan (averaging 150 and up to 300 years, with some even longer estimates), and adults appear to be less climate-sensitive than seedlings. Without severe fires, current healthy adult stands could persist into the foreseeable future, even if they are no longer reproductively viable, and support cultural and biodiversity values, genetics conservation, and ecosystem structure and function.

3. Recognize the importance of facilitated dispersal

Projected climatically suitable habitat loss under realistic climate scenarios, and western Joshua tree's low natural dispersal indicate the likely necessity of facilitated dispersal (assisted migration). The

draft plan discounts facilitated dispersal as prohibitively costly and logistically complex, in favor of prioritizing projected refugia. However, under more realistic emissions scenarios, there will likely be little remaining suitable habitat within the current distribution. The draft plan cites Ricciardi and Simberloff (2009) on ecologists' concerns around assisted migration. However, since this publication, the field of climate change ecology has developed considerably. Assisted migration is now a frequently recommended strategy by scientists for climate-adaptive management of high climate-risk species (McLaughlin et al. 2022a). In collaboration with tribal partners, it is urgent to begin the process of facilitated dispersal. Experimental plantings in newly climatically suitable habitat could offer opportunities to refine facilitated dispersal strategies, validate projected future habitat suitability, and support genetic rescue for declining populations (e.g. McLaughlin et al. 2022b).

4. Prioritize research on groundwater

Western Joshua tree is identified as a phreatophyte (Lichvar and Dixon 2007), however there is little research on the species' use of groundwater, whether groundwater can buffer climate change impacts (groundwater refugia), or sustainable levels of extraction to allow for continued support of this groundwater dependent ecosystem. If groundwater is buffering western Joshua trees from the impacts of climate, development impacts could include both direct take and increasing groundwater loss.

5. Expand effectiveness criteria

Habitat suitability analyses provide our best understanding of potential future ranges of western Joshua tree, but these predictions may be modified in light of actual range shifts, dieback, survival and recruitment, or on updated understanding of western Joshua tree physiology, and/or local effects of future climate change (as noted in Section 5.3.1, Effectiveness Criteria for Conservation). Therefore, the plan should include provisions for monitoring demographic trends, enabling an adaptive management response, with regularly updated critical thresholds.

6. Prioritize recruitment and nurse trees

Given the value of nurse plants for regenerating western Joshua trees (WJTCP report page p. 5-10), mature trees or other plants that act as nurse plants for recruits should be prioritized. Suites of adult nurse trees and recruits also should have a higher buffer of protection than trees that are not nursing seedling recruitment. In addition, healthy juveniles should be prioritized similarly to adult trees since they may have already been selected for success in a changing climate, and since they represent the future of the species.

The draft plan is an important step toward incorporating climate change into conservation planning for western Joshua tree. In closing, we emphasize the importance of incorporating more realistic emissions scenarios, the need to begin facilitated dispersal and the potential value of stands in projected areas of loss. We also recommend increasing research on groundwater interactions with climate change, expanding the effectiveness criteria to include regular monitoring, and prioritizing recruitment and nurse trees.

Respectfully,

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